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CALIBRATION DEVICE, MEDICAL IMAGE CAPTURING SYSTEM, AND CALIBRATION METHOD

Abstract

Provided are a calibration device, a medical image capturing system, and a calibration method that can easily acquire calibration data of a photon-counting detector. A calibration device is a calibration device that is used to acquire calibration data of a photon-counting detector that outputs an electrical signal corresponding to a photon energy of incident radiation, and includes: a holding portion that holds a first base substance and a second base substance having a larger attenuation coefficient for the radiation than the first base substance; and a moving mechanism that moves the holding portion in a body axis direction of a subject irradiated with the radiation to move each of the first base substance and the second base substance between a position within an irradiation field and a position outside the irradiation field.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority under 35 U.S.C § 119 from Japanese Patent Application No. 2024-035388, filed on Mar. 7, 2024, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

Technical Field

[0002] The present disclosure relates to a calibration device, a medical image capturing system, and a calibration method.

Related Art

[0003] A photon-counting computed tomography (PCCT) apparatus comprising a photon-counting detector that is a detector adopting a photon-counting method is known. Since the photon-counting detector can measure a photon energy, which is an energy of an incident radiation photon, the PCCT apparatus can obtain a medical image in which substances having different compositions are discriminated, for example, a medical image in which an iodine contrast medium used in angiography and calcified plaque in a blood vessel are discriminated. In order to obtain a medical image in which substances are discriminated, the detector is calibrated. Therefore, for combinations of a plurality of base substances, which are substances having known compositions and thicknesses, a relationship between an output measured by the photon-counting detector and a photon energy is acquired in advance as calibration data for each detector element.

[0004] JP2013-146480A discloses acquiring calibration data from each of stepped phantoms made of acrylic and aluminum.

[0005] In a case where the stepped phantom described in JP2013-146480A is applied to a large irradiation field of about 50 cm, the phantom becomes heavy and difficult to handle, and it takes time to acquire calibration data of the photon-counting detector, which results in time and effort to acquire the calibration data.

SUMMARY

[0006] The present disclosure has been made in consideration of the above circumstances, and an object of the present disclosure is to provide a calibration device, a medical image capturing system, and a calibration method that can easily acquire calibration data of a photon-counting detector.

[0007] In order to achieve the above object, a first aspect of the present disclosure provides a calibration device that is used to acquire calibration data of a photon-counting detector that outputs an electrical signal corresponding to a photon energy of incident radiation, the calibration device comprising: a holding portion that holds a first base substance and a second base substance having a larger attenuation coefficient for the radiation than the first base substance; and a moving mechanism that moves the holding portion in a body axis direction of a subject irradiated with the radiation to move each of the first base substance and the second base substance between a position within an irradiation field and a position outside the irradiation field.

[0008] A second aspect provides the calibration device according to the first aspect, in which the moving mechanism linearly moves the holding portion to move each of the first base substance and the second base substance between a position within the irradiation field and a position outside the irradiation field.

[0009] A third aspect provides the calibration device according to the first aspect, in which the holding portion includes a first holding member that holds the first base substance and a second

holding member that holds the second base substance.
[0010] A fourth aspect provides the calibration device according to the third aspect, in which the holding portion includes a plurality of the first holding members and a plurality of the second holding members.
[0011] A fifth aspect provides the calibration device according to the third aspect, in which the first base substance has a plurality of measurement regions having different thicknesses in a direction in which the radiation is transmitted, and the moving mechanism moves the first base substance to a position within the irradiation field for each of the measurement regions.
[0012] A sixth aspect provides the calibration device according to the third aspect, in which the second base substance has a plurality of measurement regions having different thicknesses in a direction in which the radiation is transmitted, and the moving mechanism moves the second base substance to a position within the irradiation field for each measurement region.
[0013] A seventh aspect provides the calibration device according to the first aspect, in which each of the first base substance and the second base substance has a plurality of divided parts, and the moving mechanism moves the first base substance and the second base substance to a position within the irradiation field for each of the parts.
[0014] An eighth aspect provides the calibration device according to the first aspect, further comprising: a controller that performs control of causing the moving mechanism to move the first base substance and the second base substance to a position within the irradiation field according to a combination of the first base substance and the second base substance.
[0015] In order to achieve the above object, a ninth aspect of the present disclosure provides a medical image capturing system comprising: a radiation source; a photon-counting detector that outputs an electrical signal corresponding to a photon energy of radiation emitted from the radiation source; and the calibration device described in the present disclosure.
[0016] In order to achieve the above object, a tenth aspect of the present disclosure provides a calibration method of a photon-counting detector that outputs an electrical signal corresponding to a photon energy of incident radiation, the calibration method comprising: holding, by a holding portion, a first base substance and a second base substance having a larger attenuation coefficient for the radiation than the first base substance; moving, by a moving mechanism, the holding portion in a body axis direction of a subject irradiated with the radiation to move each of the first base substance and the second base substance between a position within an irradiation field and a position outside the irradiation field; and acquiring a plurality of pieces of calibration data obtained by varying a combination of the first base substance and the second base substance inserted into the irradiation field.
[0017] According to the present disclosure, it is possible to easily acquire calibration data of a photon-counting detector.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0018] FIG. **1** is a configuration diagram showing an example of an overall configuration of a medical image capturing system according to an embodiment.
[0019] FIG. **2** is a diagram for describing an example of a calibration method of a detector panel, which is a photon-counting detector.
[0020] FIG. **3** is a configuration diagram showing an example of a configuration of a calibration device of the embodiment.
[0021] FIG. **4**A is a diagram for describing movement of a first base substance.
[0022] FIG. **4**B is a diagram for describing movement of a second base substance.
[0023] FIG. **5** is a configuration diagram showing an example of a configuration of a calibration

device of Modification Example 1.
[0024] FIG. **6**A is a diagram showing an example of the first base substance comprised in the calibration device of Modification Example 1.
[0025] FIG. **6**B is a diagram showing an example of the second base substance comprised in the calibration device of Modification Example 1.
[0026] FIG. **7** is a diagram showing an example of the first base substance and the second base substance comprised in a calibration device of Modification Example 2.
DETAILED DESCRIPTION
[0027] Hereinafter, embodiments of the present invention will be described in detail with reference to the drawings. The present embodiment does not limit the present invention.
[0028] First, an example of an overall configuration of a medical image capturing system according to the present embodiment will be described. FIG. **1** is a configuration diagram showing an example of an overall configuration of a medical image capturing system **10** according to the present embodiment.
[0029] As shown in FIG. **1**, the medical image capturing system **10** according to the present embodiment comprises a gantry **20**, a bed **27**, a console **30**, and a calibration device **40**. In the following description, a lateral direction in FIG. **1** is defined as an X-axis, a longitudinal direction is defined as a Y-axis, and a direction orthogonal to an XY plane is defined as a Z-axis.
[0030] The gantry **20** has an opening portion **26**, and a subject S as an imaging target is disposed in the opening portion **26** in a state of being placed on the bed **27**. The gantry **20** and the bed **27** are movable relative to each other in a Z-axis direction.
[0031] A radiation source **22** including a radiation tube **23** and a bowtie filter **24** and a detector panel **28** are disposed inside the gantry **20** in a state of facing each other with the subject S interposed therebetween. Radiation R emitted from the radiation tube **23** is shaped into a beam shape suitable for a size of the subject S by the bowtie filter **24** and irradiates the subject S. The detector panel **28** detects the radiation transmitted through the subject S and generates projection data according to a dose of the detected radiation. For example, the detector panel **28** of the present embodiment is a photon-counting detector in which a plurality of detection elements **28**P that detect a photon energy, which is an energy of a photon of the incident radiation, are arranged in an arc shape centered on a focal point **23**F of the radiation tube **23**. The detector panel **28**, which is a photon-counting detector, outputs projection data according to the photon energy.
[0032] The radiation tube **23** and the detector panel **28** are rotated around the subject S by a rotation drive unit (not shown) of the gantry **20**. The radiation irradiation from the radiation tube **23** and the detection of the radiation by the detector panel **28** are repeated with the rotation of the radiation tube **23** and the detector panel **28**, and thus projection data at various projection angles are acquired. A plurality of pieces of projection data acquired by the detector panel **28** are output to the console **30**.
[0033] The dose of the radiation emitted from the radiation tube **23**, a rotation speed of the gantry **20**, a relative movement speed between the gantry **20** and the bed **27**, and the like are set by the console **30** based on scan conditions input by a user, such as a technician.
[0034] The console **30** of the present embodiment performs control related to the acquisition of projection data, generation of a medical image, calibration of the detector panel **28** by the calibration device **40**, and the like.
[0035] The console **30** comprises a controller **32**, a storage unit **33**, an interface (I/F) unit **34**, an operation unit **36**, and a display unit **38**. The controller **32** comprises a central processing unit (CPU), a read only memory (ROM), a random access memory (RAM), and the like (all of which are not shown). For example, the console **30** of the present embodiment is a server computer.
[0036] The ROM stores in advance various programs executed by the CPU, including programs for performing the control related to the acquisition of projection data, generation of a medical image, calibration of the detector panel **28** by the calibration device **40**, and the like. The RAM

temporarily stores various types of data.
[0037] The storage unit **33** stores image data of the medical image, various other types of information, and the like. The storage unit **33** is implemented by, for example, a storage medium such as a hard disk drive (HDD), a solid state drive (SSD), and a flash memory.
[0038] The I/F unit **34** communicates various types of information with the rotation drive unit (not shown) of the gantry **20**, the radiation source **22**, the detector panel **28**, and the calibration device **40** by wired communication or wireless communication. The console **30** of the present embodiment receives the projection data and calibration data **55** from the detector panel **28** via the I/F unit **34**.
[0039] The operation unit **36** is used by the user to input scan conditions for acquiring the projection data, an instruction or various types of information related to the generation and display of an image, and the like. The operation unit **36** is not particularly limited, and examples thereof include various switches, buttons, a touch panel, a touch pen, a keyboard, and a mouse. The display unit **38** displays various types of information, medical images, and the like. The operation unit **36** and the display unit **38** may be integrated into a touch panel display. In addition, for example, the operation unit **36** may receive a voice input from the user.
[0040] The console **30** acquires a plurality of pieces of projection data from the detector panel **28** via the I/F unit **34**. The controller **32** performs a reconstruction process on the acquired plurality of pieces of projection data to generate a tomographic image of the subject S. In a case where the plurality of pieces of projection data are image data acquired while the bed **27** and the gantry **20** are moved relative to each other in the Z-axis direction, the controller **32** generates a three-dimensional image of the subject S from the plurality of pieces of projection data. In the present embodiment, the projection data obtained by the detector panel **28**, as well as the tomographic image and three-dimensional image generated by the controller **32**, are collectively referred to as “medical images”.
[0041] The calibration device **40** of the present embodiment is a device for calibrating the entire medical image capturing system **10**, mainly the detector panel **28**. In the medical image capturing system **10** comprising the detector panel **28**, which is a photon-counting detector, a photon energy spectrum related to the projection data of the subject S can be acquired. Therefore, a medical image in which substances having different compositions are discriminated or a medical image divided into a plurality of energy components can be generated. In order to obtain the medical image in which substances having different compositions are discriminated as described above, and the like, it is necessary to calibrate in advance a relationship between an output when a combination of a plurality of base substances, which are substances having known compositions and thicknesses, is measured by the detector panel **28** and the photon energy, for each of the detection elements **28**P. The calibration device **40** is a device used for the calibration. The disposition and orientation of the calibration device **40** shown in FIG. **1** are for convenience of description, and are different from the disposition and orientation of the calibration device **40** in a case where the detector panel **28** is actually configured.
[0042] Here, an example of a calibration method of the detector panel **28**, which is a photon-counting detector, will be described with reference to FIG. **2**.
[0043] For the calibration of the detector panel **28**, which is a photon-counting detector, a plurality of base substances having known compositions and thicknesses are used. In an example of the calibration shown in FIG. **2**, two types of base substances, that is, a first base substance **53_1** and a second base substance **53_2**, are used. The first base substance **53_1** and the second base substance **53_2** have different attenuation coefficients for radiation, and in the present embodiment, the second base substance **53_2** has a larger attenuation coefficient than the first base substance **53_1**. For example, examples of the first base substance **53_1** include acrylic, and examples of the second base substance **53_2** include aluminum having a larger attenuation coefficient than acrylic.
[0044] In the example shown in FIG. **2**, by combining two first base substances **53_1** having the same thickness and two second base substances **53_2** having the same thickness, the calibration data **55** is obtained for each of a plurality of combinations **54** of the first base substances **53_1** and

the second base substances **53_2** having different thicknesses in a transmission direction of the radiation R. For example, in a case where the first base substance **53_1** has J types of thicknesses and the second base substance **53_2** has K types of thicknesses, J×K pieces of calibration data **55** are obtained by J×K types of combinations **54** of base substances.

[0045] Specifically, in the example shown in FIG. **2**, assuming that a case where the first base substance **53_1** is not provided in an irradiation field RF is a case where the thickness of the first base substance **53_1** is “0 (zero)”, there are J=3 types of thicknesses of the first base substance **53_1**. Similarly, assuming that a case where the second base substance **53_2** is not provided in the irradiation field RF is a case where the thickness of the second base substance **53_2** is “0 (zero)”, there are K=3 types of thicknesses of the second base substance **53_2**. Therefore, in this case, there are 3×3=9 types of combinations of the base substances. In FIG. **2**, “Air” corresponds to a case where neither the first base substance **53_1** nor the second base substance **53_2** is provided in the irradiation field RF, that is, a case where each of the first base substance **53_1** and the second base substance **53_2** has a thickness of “0 (zero)”.

[0046] Each of the nine types of combinations **54** is inserted into the irradiation field RF of the radiation R and irradiated with the radiation R from the radiation source **22**, and the radiation R transmitted through the combination **54** is detected by the detector panel **28**, so that a photon energy spectrum is acquired as the calibration data **55** for each combination **54**. The nine types of calibration data **55** acquired in this way are output to the console **30**, are stored in the storage unit **33** of the console **30**, and are used for calibration of the projection data of the subject S.

[0047] A configuration of the calibration device **40** of the present embodiment used for such calibration will be described with reference to FIGS. **1** and **3**. The calibration device **40** of the present embodiment is configured as a carriage **58** having a housing **58_1** and a plurality of wheels **58_2**, and comprises a controller **42**, a storage unit **43**, and an I/F unit **44** in the housing **58_1**.

[0048] The controller **42** comprises a CPU, a ROM, a RAM, and the like (all of which are not shown). The ROM stores in advance various programs executed by the CPU, including programs for performing control related to acquisition of the calibration data **55**. The RAM temporarily stores various types of data. The storage unit **43** stores various types of information and the like. The storage unit **43** is implemented by, for example, a storage medium such as an HDD, an SSD, and a flash memory. The I/F unit **44** communicates various types of information with the console **30** by wired communication or wireless communication. Specifically, the I/F unit **44** receives information related to the control for acquiring the calibration data **55** from the console **30**.

[0049] Furthermore, the calibration device **40** comprises a holding portion **52** and a moving mechanism **50**. As shown in FIG. **3**, the holding portion **52** includes a plurality of first holding members **52_1** and a plurality of second holding members **52_2**. The calibration device **40** comprises four first base substances **53_1** (**53_11** to **53_14**), and the first holding members **52_1** respectively hold the first base substances **53_11** to **53_14**. As an example, in the present embodiment, as shown in FIG. **4**A, a pair of the first holding members **52_1** respectively hold a pair of opposite sides extending in the Z-axis direction of the first base substance **53_1** of which a surface (XZ surface in FIG. **4**A) intersecting an irradiation direction of the radiation R is rectangular. In addition, the calibration device **40** comprises four second base substances **53_2** (**53_21** to **53_24**), and the second holding member **52_2** respectively hold the second base substances **53_21** to **53_24**. As an example, in the present embodiment, as shown in FIG. **4**B, a pair of the second holding members **52_2** respectively hold a pair of opposite sides extending in the Z-axis direction of the second base substance **53_2** of which a surface (XZ surface in FIG. **4**B) intersecting the irradiation direction of the radiation R is rectangular. As shown in FIGS. **4**A and **4**B, one ends of the first holding member **52_1** and the second holding member **52_2** are connected to the moving mechanism **50** provided in the housing **58_1** of the carriage **58**.

[0050] The moving mechanism **50** moves the holding portion **52** in a body axis direction of the subject S irradiated with the radiation R under the control of the controller **42** to move each of the first base substance **53_1** and the second base substance **53_2** between a position within the irradiation field RF and a position outside the irradiation field RF. In the present embodiment, as shown in FIGS. **3**, **4**A, and **4**B, the moving mechanism **50** linearly moves each of the first holding member **52_1** and the second holding member **52_2** in an a direction along the Z-axis to move each of the first base substance **53_1** and the second base substance **53_2** between the position within the irradiation field RF and the position outside the irradiation field RF. FIG. **3** shows a state in which the first base substances **53_11** and **53_12** and the second base substances **53_23** and **53_24** are disposed at positions within the irradiation field RF and the first base substances **53_13** and **53_14** and the second base substances **53_21** and **53_22** are disposed at positions outside the irradiation field RF.

[0051] As the moving mechanism **50** capable of linearly moving the first holding member **52_1** and the second holding member **52_2** in the a direction, for example, a linear actuator or the like can be used.

[0052] Next, the calibration method of the detector panel **28** using the moving mechanism **50** of the present embodiment will be described. The calibration described here is performed in a state in which the gantry **20** is not rotated.

[0053] First, a person in charge of the calibration moves the carriage **58** to dispose the calibration device **40** at a predetermined position in front of the gantry **20**. The predetermined position is a position at which the moving mechanism **50** can move the holding portion **52** to move the first base substance **53_1** and the second base substance **53_2** between the position within the irradiation field RF and the position outside the irradiation field RF of the radiation R.

[0054] After the calibration device **40** is disposed, the person in charge gives an instruction to perform the calibration via the operation unit **36** of the console **30**. In a case where the instruction is received, the controller **32** of the console **30** instructs the calibration device **40** to acquire the calibration data **55** via the I/F unit **34**.

[0055] In a case where the calibration device **40** receives the instruction to acquire the calibration data **55** from the console **30** via the I/F unit **44**, the controller **42** instructs the moving mechanism **50** to move the first base substance **53_1** and the second base substance **53_2**.

[0056] As described above, the calibration device of the present embodiment comprises the four first base substances **53_1** (**53_11** to **53_14**) and the four second base substances **53_2** (**53_21** to **53_24**). Therefore, including the case where the thickness of each of the first base substance **53_1** and the second base substance **53_2** is “0 (zero)”, 5×5=25 types of combinations **54** of the first base substances **53_1** and the second base substances **53_2** are obtained.

[0057] The controller **42** controls the moving mechanism **50** such that the first base substances **53_1** and the second base substances **53_2** are sequentially placed within the irradiation field RF according to the 25 types of combinations **54**. The moving mechanism **50** moves at least one of the first holding member **52_1** or the second holding member **52_2** in the a direction under the control of the controller **42** to move the first base substances **53_1** and the second base substances **53_2** to the positions within the irradiation field RF of the opening portion **26** of the gantry **20**.

[0058] In a case where the first base substances **53_1** and the second base substances **53_2** are moved to a state corresponding to any of the 25 types of combinations **54**, the controller **42** transmits arrangement completion information indicating that the arrangement of the first base substances **53_1** and the second base substances **53_2** is completed, to the console **30** via the I/F unit **44**. In a case where the console **30** receives the arrangement completion information from the calibration device **40** via the I/F unit **34**, the console **30** instructs the radiation source **22** to perform the irradiation of the radiation R to acquire the calibration data **55**. A radiation source controller (not shown) of the radiation source **22** irradiates the combination **54** disposed in the opening portion **26** of the gantry **20** with the radiation R from the radiation tube **23** in response to the irradiation instruction. The radiation R transmitted through the combination **54** is detected by the detector panel **28**, and the calibration data **55** is acquired and output to the console **30**.

[0059] The console **30** stores the calibration data **55** acquired from the detector panel **28** in the storage unit **33** in association with the type of the combination **54** used to acquire the calibration data **55**.
[0060] In the medical image capturing system **10** according to the present embodiment, as described above, the 25 types of calibration data **55** are obtained by repeating the movement of the first base substances **53_1** and the second base substances **53_2** by the moving mechanism **50** of the calibration device **40**, the emission of the radiation R from the radiation tube **23**, and the detection of the radiation R by the detector panel **28**. The controller **32** performs calibration of the detector panel **28** using the 25 types of calibration data **55**. It should be noted that the calibration method of the detector panel **28** using the calibration data **55** is not limited, and a known method can be applied.
[0061] As described above, with the medical image capturing system **10** of the present embodiment, the detector panel **28** can be calibrated by using the calibration device **40**. The technology of the present disclosure is not limited to the above-described embodiment, and may be modified as in Modification Examples 1 and 2 described below.
Modification Example 1
[0062] In the above-described embodiment, in order to vary the thickness of each of the first base substance **53_1** and the second base substance **53_2** in the transmission direction of the radiation R, a form in which the calibration device **40** comprises the four first base substances **53_1** and the four second base substances **53_2** has been described. However, the numbers of the first base substance **53_1** and the second base substance **53_2** comprised in the calibration device **40** and the like are not limited to the above-described embodiment.
[0063] FIG. **5** is a configuration diagram of the calibration device **40** of the present modification example. The calibration device **40** of the present modification example is different from the calibration device **40** (see FIG. **3**) of the above-described embodiment in that the calibration device **40** of the present modification example comprises one first base substance **53_1** and one second base substance **53_2**.
[0064] As shown in FIG. **6**A, the first base substance **53_1** of the present modification example has five measurement regions **60_1** having different thicknesses in a direction in which the radiation R is transmitted. The moving mechanism **50** of the present modification example moves the first holding member **52_1** in the a direction to move the first base substance **53_1** to a position within the irradiation field RF for each measurement region **60_1**.
[0065] In addition, as shown in FIG. **6**B, the second base substance **53_2** of the present modification example has five measurement regions **60_2** having different thicknesses in the direction in which the radiation R is transmitted. The moving mechanism **50** of the present modification example moves the second holding member **52_2** in the a direction to move the second base substance **53_2** to a position within the irradiation field RF for each measurement region **60_2**.
[0066] As described above, the calibration device **40** of the present modification example has the five measurement regions **60_1** in which the first base substance **53_1** has different thicknesses in the transmission direction of the radiation R, and has the five measurement regions **60_2** in which the second base substance **53_2** has different thicknesses in the transmission direction of the radiation R. Accordingly, with the calibration device **40** of the present modification example, it is possible to acquire 25 types of calibration data **55** in the same manner as the calibration device **40** (see FIG. **3**) of the above-described embodiment.
[0067] Therefore, with the calibration device **40** of the present modification example, the numbers of the first holding members **52_1** and the second holding members **52_2** can be reduced, and the moving mechanism **50** can be reduced in size. For example, in a case where the linear actuator is provided for each of the first holding member **52_1** and the second holding member **52_2**, the number of linear actuators comprised in the moving mechanism **50** can be suppressed.

Modification Example 2

[0068] Each of the first base substance **53_1** and the second base substance **53_2** comprised in the calibration device **40** may have a plurality of divided parts. In an example shown in FIG. **7**, the first base substance **53_1** has two divided parts, that is, a normal region **53_1***a* and a reference region **53_1***b*. In addition, the second base substance **53_2** has two divided parts, that is, a normal region **53_2***a* and a reference region **53_2***b*. The moving mechanism **50** moves the first base substance **53_1** and the second base substance **53_2** to a position within the irradiation field RF for each of the parts. In FIG. **7**, one first base substance **53_1** and one second base substance **53_2** are shown, but a plurality of first base substances **53_1** and a plurality of second base substances **53_2** may be provided as in the above-described embodiment.

[0069] The reference regions **53_1***b* and **53_2***b* are regions corresponding to an irradiation field RF of the radiation R irradiating a reference detector **28**R. In the present modification example, a predetermined number of the detection elements **28**P at one end portion of the detector panel **28** are used as the reference detectors **28**R. The reference detector **28**R detects the radiation R that is not transmitted through the subject S and outputs a detection result as reference data. The controller **32** of the console **30** calibrates intensity levels of the radiation obtained by the other detection elements **28**P based on an intensity of the radiation obtained by the reference data, and generates a tomographic image using a signal after the calibration.

[0070] In the calibration device **40** of the present modification example, each of the normal region **53_1***a* and the reference region **53_1***b* can be moved between a position within the irradiation field RF and a position outside the irradiation field RF by the moving mechanism **50**. For example, a form in which the reference region **53_1***b* is held by a first holding member **52_1** different from the first holding member **52_1** that holds the normal region **53_1***a* may be adopted, and each first holding member **52_1** may be moved linearly by the moving mechanism **50**. Similarly, a form in which the reference region **53_2***b* is held by a second holding member **52_2** different from the second holding member **52_2** that holds the normal region **53_2***a* may be adopted, and each second holding member **52_2** may be moved linearly by the moving mechanism **50**.

[0071] In addition, a specific method of acquiring the calibration data **55** and the reference data in this case is not limited, but may be, for example, the following method. First, the controller **42** of the calibration device **40** causes the moving mechanism **50** to position both the normal region **53_1***a* and the reference region **53_1***b* within the irradiation field RF. The calibration data **55** is acquired in a state in which both the normal region **53_1***a* and the reference region **53_1***b* are positioned within the irradiation field RF. Next, the controller **42** causes the moving mechanism **50** to move the normal region **53_1***a* to a position outside the irradiation field RF. Reference data is acquired in a state in which only the reference region **53_1***b* is positioned within the irradiation field RF. A series of processes for acquiring the calibration data **55** and the reference data is performed for each combination **54**.

[0072] As described above, according to the present modification example, the reference data can also be easily acquired by the calibration device **40**.

[0073] In the present embodiment, the form in which one end portion (right end portion in FIG. **7**) of the detector panel **28** is the reference detector **28**R has been described, but the present invention is not limited to this form. For example, both end portions of the detector panel **28** may be used as the reference detector **28**R. In this case, the first base substance **53_1** has three divided parts, that is, one normal region **53_1***a* and two reference regions **53_1***b*. In addition, the second base substance **53_2** has three divided parts, that is, one normal region **53_2***a* and two reference regions **53_2***b*.

[0074] As described above, the calibration device **40** of the above-described embodiment and each modification example is a calibration device used to acquire the calibration data **55** of the photon-counting detector panel **28** that outputs an electrical signal corresponding to the photon energy of the incident radiation R. The calibration device **40** comprises the holding portion **52** that holds the first base substance **53_1** and the second base substance **53_2** having a larger attenuation coefficient for the radiation R than the first base substance **53_1**. In addition, the calibration device **40** comprises the moving mechanism **50** that moves the holding portion **52** in the body axis direction of the subject S irradiated with the radiation R to move each of the first base substance **53_1** and the second base substance **53_2** between a position within the irradiation field RF and a position outside the irradiation field RF.

[0075] For example, in a case of a calibration device using a stepped phantom as described in JP2013-146480A, unlike the calibration device **40** according to the above-described embodiment and each modification example, the number of steps increases according to the number of combinations of base substances required, and thus the entire calibration device becomes larger and heavier. Contrary to this, in the calibration device **40** of the above-described embodiment and each modification example, since the moving mechanism **50** moves the first base substance **53_1** and the second base substance **53_2** as described above, the entire calibration device **40** can be reduced in size. Therefore, with the calibration device **40** of the above-described embodiment and each modification example, the calibration data of the photon-counting detector can be easily acquired.

[0076] It should be noted that the configurations and operations of the medical image capturing system **10**, the console **30**, the calibration device **40**, and the like described in the above-described embodiment and each modification example are merely examples, and it goes without saying that the configurations and operations can be changed according to the situation without departing from the spirit of the present invention. In addition, it goes without saying that the above-described embodiments may be appropriately combined.

[0077] Further, the following supplementary notes will be disclosed with respect to the above-described embodiments.

Supplementary Note 1

[0078] A calibration device that is used to acquire calibration data of a photon-counting detector that outputs an electrical signal corresponding to a photon energy of incident radiation, the calibration device comprising: a holding portion that holds a first base substance and a second base substance having a larger attenuation coefficient for the radiation than the first base substance; and a moving mechanism that moves the holding portion in a body axis direction of a subject irradiated with the radiation to move each of the first base substance and the second base substance between a position within an irradiation field and a position outside the irradiation field.

Supplementary Note 2

[0079] The calibration device according to supplementary note 1, in which the moving mechanism linearly moves the holding portion to move each of the first base substance and the second base substance between a position within the irradiation field and a position outside the irradiation field.

Supplementary Note 3

[0080] The calibration device according to supplementary note 1 or 2, in which the holding portion includes a first holding member that holds the first base substance and a second holding member that holds the second base substance.

Supplementary Note 4

[0081] The calibration device according to supplementary note 3, in which the holding portion includes a plurality of the first holding members and a plurality of the second holding members.

Supplementary Note 5

[0082] The calibration device according to any one of supplementary notes 1 to 4, in which the first base substance has a plurality of measurement regions having different thicknesses in a direction in which the radiation is transmitted, and the moving mechanism moves the first base substance to a position within the irradiation field for each of the measurement regions.

Supplementary Note 6

[0083] The calibration device according to any one of supplementary notes 1 to 4, in which the second base substance has a plurality of measurement regions having different thicknesses in a direction in which the radiation is transmitted, and the moving mechanism moves the second base substance to a position within the irradiation field for each measurement region.

Supplementary Note 7

[0084] The calibration device according to any one of supplementary notes 1 to 6, in which each of the first base substance and the second base substance has a plurality of divided parts, and the moving mechanism moves the first base substance and the second base substance to a position within the irradiation field for each of the parts.

Supplementary Note 8

[0085] The calibration device according to any one of supplementary notes 1 to 7, further comprising: a controller that performs control of causing the moving mechanism to move the first base substance and the second base substance to a position within the irradiation field according to a combination of the first base substance and the second base substance.

Supplementary Note 9

[0086] A medical image capturing system comprising: a radiation source; a photon-counting detector that outputs an electrical signal corresponding to a photon energy of radiation emitted from the radiation source; and the calibration device according to any one of supplementary notes 1 to 8.

Supplementary Note 10

[0087] A calibration method of a photon-counting detector that outputs an electrical signal corresponding to a photon energy of incident radiation, the calibration method comprising: holding, by a holding portion, a first base substance and a second base substance having a larger attenuation coefficient for the radiation than the first base substance; moving, by a moving mechanism, the holding portion in a body axis direction of a subject irradiated with the radiation to move each of the first base substance and the second base substance between a position within an irradiation field and a position outside the irradiation field; and acquiring a plurality of pieces of calibration data obtained by varying a combination of the first base substance and the second base substance inserted into the irradiation field.

EXPLANATION OF REFERENCES

[0088] **10**: medical image capturing system [0089] **20**: gantry [0090] **22**: radiation source [0091] **23**: radiation tube, **23**F: focal point [0092] **24**: bowtie filter [0093] **26**: opening portion [0094] **27**: bed [0095] **28**: detector panel, **28**P: detection element, **28**R: reference detector [0096] **30**: console [0097] **32**, **42**: controller [0098] **33**, **43**: storage unit [0099] **34**, **44**: I/F unit [0100] **36**: operation unit [0101] **38**: display unit [0102] **40**: calibration device [0103] **50**: moving mechanism [0104] **52**: holding portion, **52_1**: first holding member, **52_2**: second holding member [0105] **53_1**, **53_11** to **53_14**: first base substance, **53_1***a*: normal region, **53_1***b*: reference region, **53_2**, **53_21** to **53_24**: second base substance, **53_2***a*: normal region, **53_2***b*: reference region [0106] **54**: combination [0107] **55**: calibration data [0108] **58**: carriage, **58_1**: housing, **58_2**: wheel [0109] **60_1**, **60_2**: measurement region [0110] R: radiation, RF: irradiation field [0111] S: subject [0112] α: direction

Claims

1. A calibration device that is used to acquire calibration data of a photon-counting detector that outputs an electrical signal corresponding to a photon energy of incident radiation, the calibration device comprising: a holding portion that holds a first base substance and a second base substance having a larger attenuation coefficient for the radiation than the first base substance; and a moving mechanism that moves the holding portion in a body axis direction of a subject irradiated with the radiation to move each of the first base substance and the second base substance between a position within an irradiation field and a position outside the irradiation field.

2. The calibration device according to claim 1, wherein the moving mechanism linearly moves the holding portion to move each of the first base substance and the second base substance between a position within the irradiation field and a position outside the irradiation field.

3. The calibration device according to claim 1, wherein the holding portion includes a first holding member that holds the first base substance and a second holding member that holds the second base substance.
4. The calibration device according to claim 3, wherein the holding portion includes a plurality of the first holding members and a plurality of the second holding members.
5. The calibration device according to claim 3, wherein: the first base substance has a plurality of measurement regions having different thicknesses in a direction in which the radiation is transmitted, and the moving mechanism moves the first base substance to a position within the irradiation field for each of the measurement regions.
6. The calibration device according to claim 3, wherein: the second base substance has a plurality of measurement regions having different thicknesses in a direction in which the radiation is transmitted, and the moving mechanism moves the second base substance to a position within the irradiation field for each measurement region.
7. The calibration device according to claim 1, wherein: each of the first base substance and the second base substance has a plurality of divided parts, and the moving mechanism moves the first base substance and the second base substance to a position within the irradiation field for each of the parts.
8. The calibration device according to claim 1, further comprising a controller that performs control of causing the moving mechanism to move the first base substance and the second base substance to a position within the irradiation field according to a combination of the first base substance and the second base substance.
9. A medical image capturing system comprising: a radiation source; a photon-counting detector that outputs an electrical signal corresponding to a photon energy of radiation emitted from the radiation source; and the calibration device according to claim 1.
10. A calibration method of a photon-counting detector that outputs an electrical signal corresponding to a photon energy of incident radiation, the calibration method comprising: holding, by a holding portion, a first base substance and a second base substance having a larger attenuation coefficient for the radiation than the first base substance; moving, by a moving mechanism, the holding portion in a body axis direction of a subject irradiated with the radiation to move each of the first base substance and the second base substance between a position within an irradiation field and a position outside the irradiation field; and acquiring a plurality of pieces of calibration data obtained by varying a combination of the first base substance and the second base substance inserted into the irradiation field.